higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 3

(12030333)

11 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

NOT

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly cross out ALL work you do not want to be marked.
5 Start each question on a NEW page.
6. Rule off across the page on completion of each question.
7. ALL the abbreviations and symbols must comply with the latest National Building Regulations and all relevant SANS (SABS) codes.
8. Write neatly and legibly.

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QUESTION 1: HIPPED ROOFS

1.1 A roof can be either 'rationally designed' or 'deemed to satisfy'. Indicate whether the following will be RATIONALLY DESIGNED or DEEMED TO SATISFY. Write only 'Rationally designed' or 'Deemed to satisfy' next to the question number (1.1.1–1.1.6) in the ANSWER BOOK.

1.1.1 A gabled roof for a small dwelling

1.1.2 A roof with a 10-meter span

1.1.3 A roof with hipped ends

1.1.4 A roof design based on the carpenter's experience

1.1.5 A roof where the manufacturing process is based on scientific calculations

1.1.6 A roof designed using the SANS load tables

(6 × 1) (6)

1.2 A carpenter must consult the 'roof layout' plan before erecting the roof.

List THREE important details found on this plan which will enable him to erect the roof correctly. (3)

1.3 Explain how the carpenter could remedy the following problems before fixing the wall plates:

1.3.1 The building is 200 mm longer than it should be.

1.3.2 The tops of the walls are uneven.

1.3.3 The building is 150 mm narrower than it should be.

(3 × 2) (6)

1.4 Describe a method of fixing the wall plates when the mortar between the bricks is too hard to accept nails. (3)

1.5 What is the meaning of the term *self-bracing?* (2)

1.6 Name the additional truss used to support the trusses when there is no supporting wall. (2)

1.7 Name the THREE trusses used when constructing the hipped end of a roof. (3)

1.8 List FIVE factors that can affect the stability of a roof. (5)

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QUESTION 2: ERECTING AN L-SHAPED ROOF

2.1 Proper on-site preparation is essential to ensure safe roof construction.

Describe FOUR ways of caring for roof trusses before they are erected. (4)

2.2 Name the simple, yet accurate handmade tool used for checking that the external load-bearing walls are the same height. (1)

2.3 Indicate whether the following statements are TRUE or FALSE. Read the statements and write 'True' or 'False' next to the question number (2.3.1–2.3.5) in the ANSWER BOOK.

2.3.1 In a mono-pitched roof only one side of the supporting walls receives wall plates.

2.3.2 A datum is a reference mark or line that measurements are taken from.

2.3.3 Jack trusses support the extra weight of the girder.

2.3.4 A 100-mm discrepancy in the size of a building is generally acceptable.

2.3.5 A ladder is by far the best option to use when fixing fascia boards.

(5 × 1) (5)

2.4 Give TWO reasons why the girder truss for an L-shaped roof must be rationally designed. (2)

2.5 45° metal hangers are fixed to the girder truss after it is in position.

2.5.1 What purpose do the 45° hangers serve?

2.5.2 Where on the girder truss are the 45° hangers fixed?

(2 × 1) (2)

2.6 What type of valley lining would necessitate the need for 'lapping'? (1)

2.7 What is the minimum overlap required for the type of valley lining referred to in Question 2.6? (1)

2.8 Choose the correct word or words from those given in brackets. Write only the word or words next to the question number (2.8.1–2.8.5) in the ANSWER BOOK.

2.8.1 A valley results from the intersection of two roof planes on (an external; an internal; the lowest) corner of a building.

2.8.2 The position of the first truss should be kept about (50 mm; 100 mm; 150 mm) from the gable wall.

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2.8.3 Runners are fixed to the (top chords; tie beams; webs) of a truss.

2.8.4 Multiple-ply girders are assembled on the ground using (screws and nails; nail plates; nails and bolts).

2.8.5 Gaps under wall plates must be filled with (grout; wedges; hoop iron) to improve stability.

(5 × 1) (5)

2.9 Below are the steps for fixing fascia. Arrange these steps in the correct order. Write only the question numbers (2.9.1–2.9.9) in the correct order in the ANSWER BOOK.

2.9.1 Tack a nail into both marks.

2.9.2 Check the fascia for straightness.

2.9.3 Mark the width of the eaves on top of the last rafter.

2.9.4 Confirm the width of the eaves.

2.9.5 Fix the fascia boards to the ends of the rafters.

2.9.6 Cut all the rafters.

2.9.7 Mark all the rafters for the plumb cuts from the gut line.

2.9.8 Tie a gut line from nail to nail.

2.9.9 Mark the width of the eaves on the top of the first rafter.

(9 × 1) (9)

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QUESTION 3: INSPECTION OF ROOF STRUCTURES

3.1 Which items of information would a roof inspector find on a truss drawing? (4)

3.2 List THREE factors the roof inspector would check when inspecting battens or purlins in a roof. (3)

3.3 Explain the difference between a *latent defect* and a *patent defect*. (2)

3.4 Give TWO examples of latent defects in a roof. (2)

3.5 Give TWO examples of patent defects in a roof. (2)

3.6 Which individual is responsible for fixing the defects referred to in QUESTION 3.4? (1)

3.7 Who is the only person entitled to sign off a rationally designed timber roof? (1)

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QUESTION 4: ROOF CLADDING

4.1 Below are the steps for tiling a roof. Arrange these steps in the correct order. Write only the question numbers (4.1.1–4.1.10) in the correct order in the ANSWER BOOK.

4.1.1 Fix the rake tiles.

4.1.2 Lay the DPC over the ridge.

4.1.3 Screw the verge counter batten to the tilting batten.

4.1.4 Mark the overhang and trim the battens.

4.1.5 Mix the mortar for the ridge tiles.

4.1.6 Start laying the tiles from the bottom right-hand side.

4.1.7 Fix the ridge tiles.

4.1.8 Establish the verge overhang.

4.1.9 Mark vertical guide lines with a chalk line.

4.1.10 Fix the tiles to the battens with 40-mm ring shank nails.

(10 × 1) (10)

4.2 Give the reason for adding pigment and bonding liquid to the mortar used for embedding the ridge tiles. (2)

4.3 SANS 10062 states that the width of the valley flashing depends on the pitch of the valley.

What is the minimum width of flashing to be used with the following valley pitches:

4.3.1 30°

4.3.2 20°

4.3.3 17,5°

(3 × 1) (3)

4.4 Name SIX factors that affect the fixing of roof tiles. (6)

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4.5 Complete the following statements by using the words given in the list below. Write only the word or words next to the question number (4.5.1–4.5.4) in the ANSWER BOOK.

roof plane; fascia; in close contact; eaves; purlins

4.5.1 On a gable roof, the ridge and the … should be parallel.

4.5.2 The sloped rectangle which forms the … should have its diagonals equal.

4.5.3 The spacing of the … depends on the type of sheeting material used.

4.5.4 White rust is caused when roofing sheets are … with each other, limiting their exposure to air.

(4 × 1) (4)
[25]

TOTAL: 100

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